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Ensuring Asset Integrity via Non-Destructive Evaluation of Mechanical Properties in Aging Infrastructure

P. Nicklas and K. Esaklul, Occidental Petroleum Corporation, Houston, Texas, USA; F. M. Haggag, M. F. Haggag, and C. K. Ellis, ABI Technology, Irvine, California, USA

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As assets age and operating conditions change assessment for continuous service and life extension necessitate review of equipment integrity and determine their fit for service conditions and set operating limits. One approach is to utilize industry's best practices as outlined in API RP 579/ASME FFE-1 Fitness-For-Service⁽¹⁾. This approach requires the use of fracture toughness of the material of construction that is not given in typical Material Test Reports (MTRs) or not available at specific operating conditions such as lower temperature Charpy Impact Energy (CVN). To enable effective analysis and reduce the degree of conservatism, Charpy Impact and/or fracture toughness data at the desired operating conditions is needed.

Since both properties are typically generated by laboratory tests that require removal of samples from multiple components, traditional destructive testing becomes less practical due to its invasive nature, downtime requirements, and the removal of materials (for machining various mechanical test specimens) from critical components. This approach is deemed impractical especially since multiple tests may be needed to determine the statistical distribution of properties of many piping or vessel segments that could be made from different heats. This led to looking for ways to obtain the required data using non-destructive means.

Accurate, nondestructive evaluation of mechanical properties such as yield strength, tensile strength, and fracture toughness is increasingly becoming critical for pipeline and pressure system integrity in upstream, midstream, and downstream oil and gas operations. As an example, an aging production platform was nearing its original design life and the current operator wished to extend its service life, by performing an in-depth engineering review of the facility and submit a life extension to the governing regulatory body for approval. During this detailed engineering review the current operator discovered that the original design for the flaring system which included the structural support of the flare boom and the process calculation for the flaring system were all adequate. However, the original design firm and operator used a unique approach by using the primary flare line for process gases, as well as primary steel support of the flare boom structure. However, no consideration for low temperature effects of rapidly expanding gases [Joule Thomson effect] during a flaring event were ever determined. To complete a comprehensive engineering analysis, the current operator needed to confirm the original materials of construction would be fit for purpose if the full capacity of the flare system was exercised during a flaring event.

Introduction

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As-built documentation of the flare boom was able to be reviewed, and according to heat/lot mapping of the flare boom fabrication, each specific Material Test Report (MTR) was able to be matched and reviewed. Unfortunately, the materials were only tested for toughness to -20F, and because part of the flare boom was fabricated to a structural welding code, AWS D1.1 and process piping normally follows ASME 31.3, there was a gap in determining the most appropriate Lowest Anticipated Service Temperature (LAST) or Minimum Design Temperature (MDT). The current operator needed to explore options to test the current materials with nondestructive methods to minimize economic impact to the facility, while providing confidence of the materials properties, specifically fracture toughness, so sound engineering judgment could be made for a life extension to this facility. The nondestructive evaluation was the only plausible solution. Both options of either taking physical samples from each heat/lot of material, or a full replacement of the flare boom, would be too obtrusive from requiring many repairs where the destructive samples were taken, to a full flare boom replacement, which was estimated to cost the operator over >\$20MM USD.

As part of an assessment strategy, a positive material identification (PMI) and a mechanical integrity evaluation using a nondestructive approach was selected to provide the best confidence to the current operator. By using PMI, the material could first be confirmed with as-built documentation, and then most importantly, the nondestructive test method could be used to verify low temperature mechanical properties, specifically fracture toughness. Several nondestructive techniques have been developed⁽²⁻⁵⁾, among them the Automated Ball Indentation[®] (ABI[®]) test, which was developed to address these constraints. This technique enables in-situ, nondestructive characterization of mechanical properties by analyzing load-depth responses generated by a controlled spherical indentation. This test replaces conventional tensile and

fracture toughness tests for ferritic steels in many applications, offering a localized, deterministic approach to material evaluation.

Commercialized in the early 1990s, [Figure \(1a\)](#), this technology has been used in nuclear, pipeline, petrochemical, automotive, and offshore applications. More recently, advances in miniaturization and portability have made this technique field-deployable for rope-access enabled inspections in offshore facilities. For pipeline and asset integrity programs, this approach enables operators to assess undocumented or degraded materials, determine the Fitness-for-Service (FFS) ⁽¹⁾ of legacy assets, and verify actual mechanical properties against regulatory or design thresholds. The technology is particularly valuable for quantifying the structural seriousness/consequences of anomalies identified via various inspection techniques such as smart pigging, ultrasonic screening methods and for confirming material grade where mill certificates are unavailable. Also, it is worth noting that mill certificates do not include fracture toughness (only chemical composition, tensile data and Charpy impact energy, and these certificates might cover several grades where requirements are met).

This paper outlines the theory and application of this nondestructive mechanical testing method, including its mechanical model, comparison to conventional testing, and recent case studies in pipeline integrity, downhole casing materials, and offshore assets. The paper also addresses the role of this technique in fracture toughness-based assessment methodologies, as increasingly required by regulatory bodies and adopted by leading operators for risk-based integrity planning. Details of the equations used to calculate the key mechanical properties from the load-depth data are published earlier in PRCI Report 52280 (April 2007)⁽²⁾.

Test Overview

This test is based on progressive indentation with intermediate partial unloadings until the desired final depth (strain) is reached, at which point the indenter is fully unloaded. The indentation force-depth data are collected continuously. The nonlinear, spherical geometry of the tungsten-carbide-indenter allows increasing strain as the indentation penetration depth is increased. The incremental values of indentation force and plastic depth associated with each partial unloading cycle are converted to incremental values of true-stress and true-plastic-strain according to elasticity/ plasticity theories. Since the test is multiaxial in nature, the stress triaxiality/intensity underneath the indenter increases with depth and can reach high values similar to or higher than those ahead of a crack tip; therefore, initiation fracture toughness values can be determined from the tests. The plastic deformation of the material surrounding the spherical indentation provides a high level of plastic constraint and no cracks would be initiated or propagated/extended, hence, no tearing modulus could be determined from ball indentation. It is considered practically nondestructive because the test leaves only a small/shallow spherical depression with a smooth surface (i.e., no sharp edges/ cracks and no stress concentration sites), and it leaves a compressive residual stress (that retards crack initiation) in the test surface area slightly larger than a shot peen.

The selected test method confirms good correlation with destructive testing results for yield strength, tensile strength, and fracture toughness. Emphasis is placed on conservative data interpretation, allowing operators to derive mechanical properties with high confidence and reduced risk. The values can be applied for re-rating transmission pipelines, process piping, re-rating of pressure vessels, and certifying downhole tubulars. The nondestructive testing is performed at room temperature to determine fracture toughness (K_{Ic}) and a conservative Reference Temperature (T_0). This T_0 is used in the master curve equation of ASTM E1921 to estimate a desired low-temperature fracture toughness value. The latter is used in Equation F.66 of API 579-1/ASME FFS-1 (2007 Fitness-For-Service)⁽¹⁾ to estimate a required low-temperature Charpy V-Notch (CVN) impact energy value without lowering the material's temperature. Case studies illustrate practical use cases such as safely increasing allowable design stress, rerating aging gas lines, and welding a nozzle in a storage tank after verifying its fracture toughness properties. These examples demonstrate how the technique supports real-world engineering decisions while reducing risk exposure and uncertainty.



Figure (1)—Automated Ball Indentation testing system on a steel pipe

Fracture Toughness Method

The Haggag Fracture Toughness Method (HFTM)^(4,5) determines the fracture toughness (K_{Jc}) value from the test on ferritic steel materials by integrating the indentation deformation energy (compression of the two surfaces of the ball indenter and the test material instead of pulling two surfaces in a destructive fracture toughness test) from the beginning of the test up to a critical indentation depth. The latter is calculated using either the critical fracture stress model or the critical fracture strain model, depending on the flow properties of the material at the test temperature. The analysis first checks the attainment of the critical fracture stress (using the mean pressure plot as a function of normalized indentation depth) before a strain value of 12% or a normalized depth of 0.6 (an example is shown in Figure (2a)). If this occurs, then the test is analyzed according to this model and all test results can be further analyzed using the fracture toughness master curve concept and a reference temperature is determined in order to evaluate the brittle behavior of the test material at low temperatures. If the critical fracture stress is not attained prior to a normalized depth of 0.6 then the specimen is analyzed according to the critical fracture strain model (an example is shown in Figure (3a and 3b) and further analysis using the fracture toughness master curve (Figure 3c) becomes invalid (or if used it will produce a very conservative reference temperature since the test material is in the ductile temperature region instead of the brittle transition region). Although the test is nondestructive, it indicates the fracture nature of the material when destructive specimens are tested (i.e., brittle if the mean pressure under the indenter meets the critical fracture stress before 12% strain or ductile behavior if the mean pressure under the ball indenter does not meet the critical fracture stress value, see Figure 3d). This capability is important when estimating CVN impact values using the appropriate empirical equations of API 579-1/ASME FFS-1, 2007, i.e. F.66, F.65, or F.60).

For ferritic steels (with yield strength of 275 to 825 MPa or 40-120 ksi) the fracture toughness (median value) versus temperature in the transition temperature region is expressed by the master curve (ASTM E-1921-97):⁽⁶⁾

$$K_{Jc}(med) = 30 + 70e^{0.019(T-T_0)} \quad \text{MPa}\sqrt{\text{m}} \quad (1)$$

Where T is the test temperature and T_0 is the reference temperature when $K_{Jc} = 100 \text{ MPa}\sqrt{\text{m}}$.

In order to obtain median dynamic fracture toughness (K_{Id}) values as a function of temperature, the ASTM Standard E1921-97 equation of the static fracture toughness (K_{Jc}) master curve can be used provided that the reference temperature is shifted to a higher value that depends on the yield strength of the test material at room temperature. It is well known that the dynamic fracture toughness curve is shifted to the right hand-side of the static fracture toughness curve by a temperature shift value depending on the room-temperature yield strength of the ferritic steel material. The median dynamic fracture toughness (K_{Id}) can be calculated from the following equation:

$$K_{Id}(med) = 30 + 70e^{0.019(T - [T_0 + T_{shift}])} \text{ MPa}\sqrt{\text{m}} \quad (2)$$

Where T is the test temperature in $^{\circ}\text{C}$ and T_0 is the reference temperature when $K_{Jc} = 100 \text{ MPa}\sqrt{\text{m}}$. The T_{shift} can be determined from the Barsom correlation and using the average yield strength measured from multiple tests at room temperature. The Barsom correlation is given by:

$$T_{shift}(^{\circ}\text{F}) = 215 - 1.5\sigma_{ys}(\text{ksi}) \quad \text{for } 36 \text{ ksi} < \sigma_{ys} < 140 \text{ ksi} \quad (3)$$

The Critical Fracture Stress (σ_f)

The σ_f value as a function of test temperature was calculated using the semi-empirical Equation 4 (given below and the fracture toughness and yield strength values of nuclear pressure vessel steel material from Oak Ridge National Laboratory (ORNL). Using Equation 4 and the ORNL data produces the tabulated values of critical fracture stress as a function of test temperature shown in Table 1 below and plotted in Figure 4.

$$\sigma_f = \sigma_y \left[1 + \ln \left(1 + 2360 \left(\frac{K_{Jc}}{\sigma_y} \right)^2 \right) \right] = 1.1 P_m \quad (4)$$

Where P_m is the mean pressure underneath the spherical indenter, "ln" is the natural logarithm, and K_{Jc} is the cleavage fracture toughness. The maximum stress ($1.1 P_m$) increases as the indenter depth increases and when it reaches the critical fracture stress value at the ABI test temperature before attaining a normalized depth of $d_t/D = 0.6$ (i.e., 12% strain), then the test is analyzed according to the critical fracture stress model and the concept of fracture toughness master curve is applicable where the reference temperature is determined from a minimum of 3 tests. Numerous tests conducted on various ferritic steels at low-test temperatures produced reference temperatures that are within 5°C from those determined from destructive fracture toughness specimens tested according to ASTM Standards E1820 and E1921.

Critical Fracture Strain Model

If the maximum stress at the indenter test temperature is not reached before a normalized indentation depth of $d/D = 0.6$ (i.e., 12% strain where "d" is the indentation chordal diameter on the sample surface and "D" is the indenter diameter), then the test is analyzed using the critical fracture strain model by integrating the indentation deformation energy (mean pressure as a function of depth) up to an empirically conservative depth value of 12% strain. Since localized cooling of a test area of an in-service structure (a pipeline or a storage tank, etc.) is not practical or safe, the reasonably conservative (within 10%) fracture toughness values obtained from a minimum of three tests, analyzed using the critical fracture strain model, can be used to determine a very conservative reference temperature (up to 70°C higher than that if the critical fracture stress model was applied on samples tested at low temperatures).

Examples of HFTM results (Interlaboratory Study): Examples of the HFTM-measured fracture toughness results on nuclear pressure vessel steel are shown in Table 1.

Table 1—Critical Fracture Stress of Ferritic Steels as a Function of Test Temperature

Test Temperature, (°C)	Critical Fracture Stress, σ_f (MPa)	Test Temperature, (°F)	Critical Fracture Stress, σ_f (ksi)
-100	2275	-148	330
-90	2280	-130	331
-80	2295	-112	333
-70	2322	-94	337
-60	2363	-76	343
-50	2417	-58	351
-40	2488	-40	361
-30	2575	-22	373
-20	2680	-4	389
-10	2804	14	407
0	2949	32	428
10	3115	50	452
20	3304	68	479

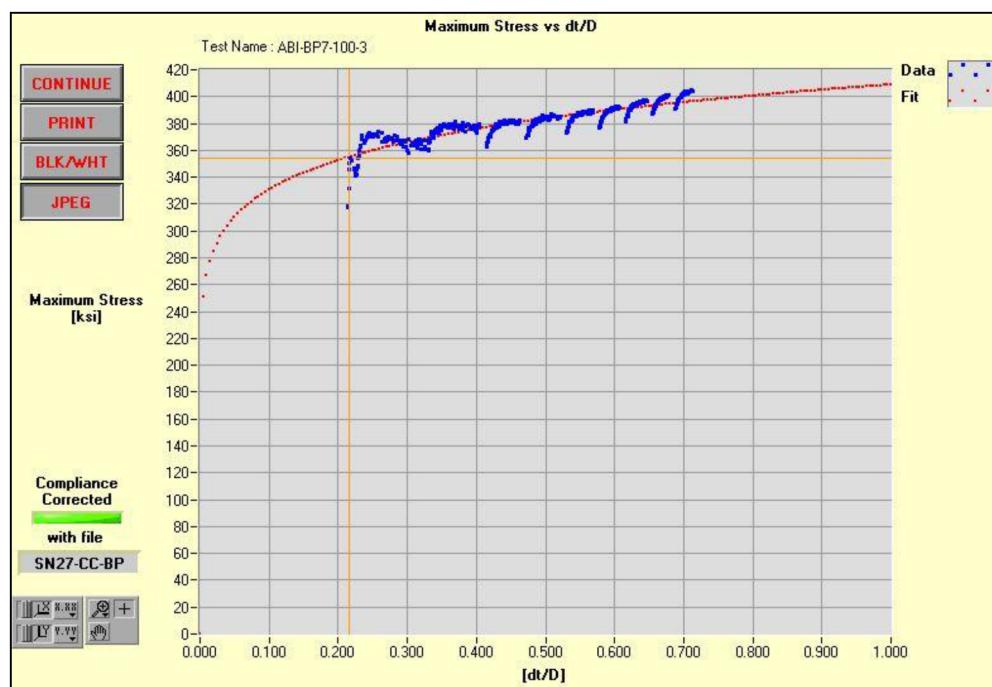


Figure (2a)—Example of the use of the critical fracture stress model to determine the critical indentation depth (corresponding to the critical fracture stress at -100°C) in order to integrate the indentation deformation energy up to the critical depth to calculate fracture toughness from the multi-axis Indentation test conducted at -100°C on steel from a storage tank.

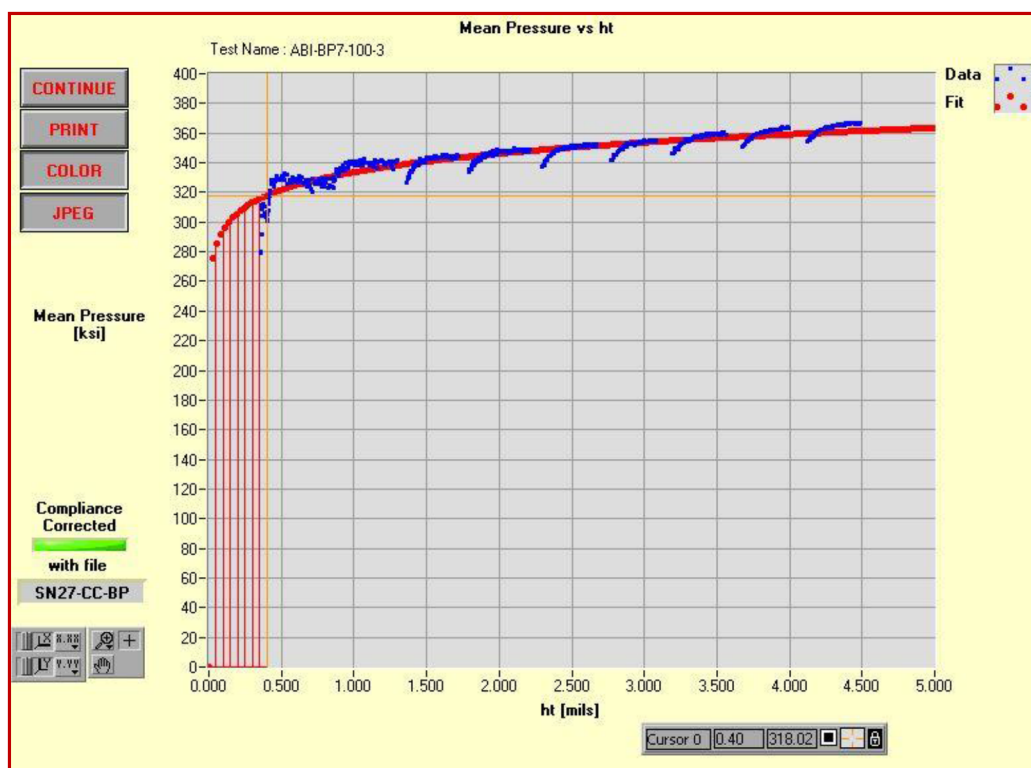


Figure (2b)—Example of the use of the critical fracture stress model to determine the temperature-dependent part of the fracture toughness by integrating the deformation energy (mean pressure under the ball indenter) versus depth (corresponding to the critical fracture stress at -100°C shown above in Fig. 1a).

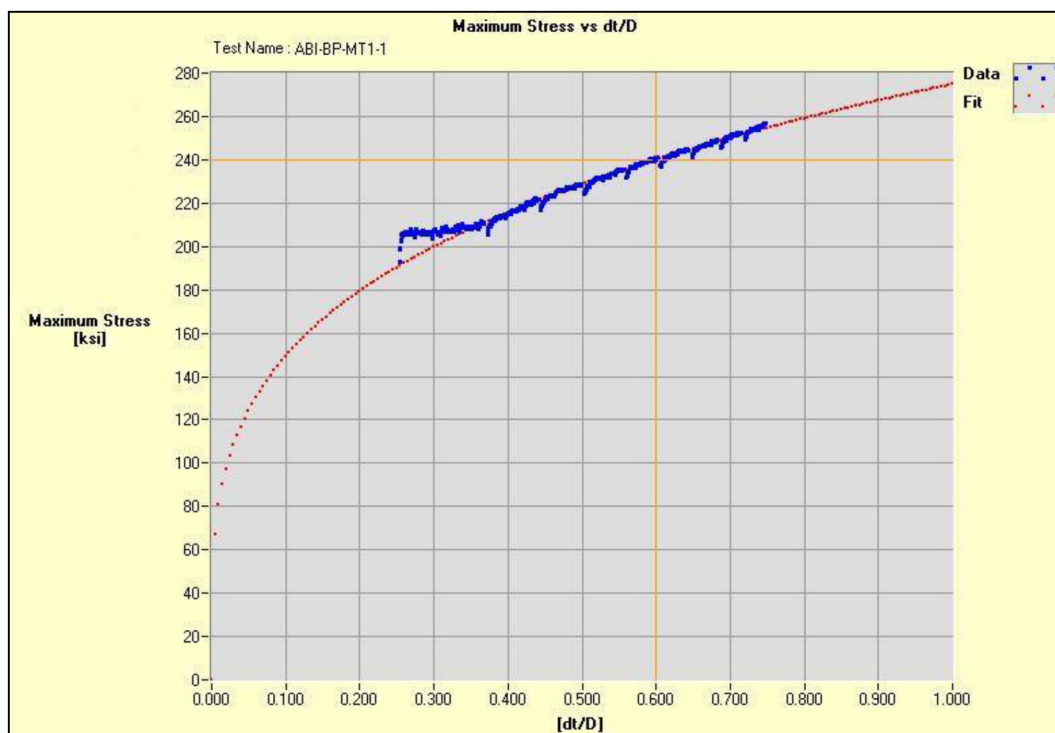


Figure (3a)—Example of the use of the critical fracture strain model when testing a ductile material at room temperature where the critical fracture stress (479 ksi at 20°C) is not reached at a normalized depth of $dt/D = 0.6$.

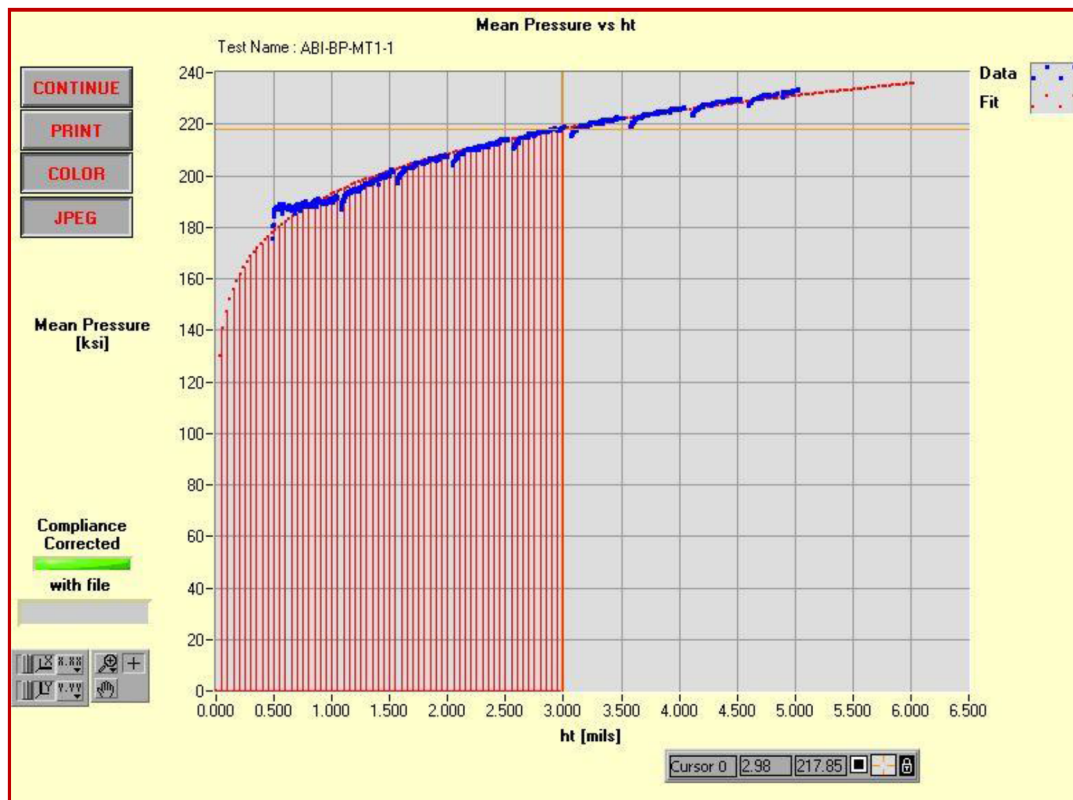


Figure (3b)—Example of the use of the critical fracture strain model to determine the temperature-dependent part of the fracture toughness by integrating the deformation energy (mean pressure under the ball indenter) versus depth (corresponding to the critical fracture strain of 12% at $dt/D = 0.6$ when testing a ductile material at room temperature as shown above in Fig. 3a).

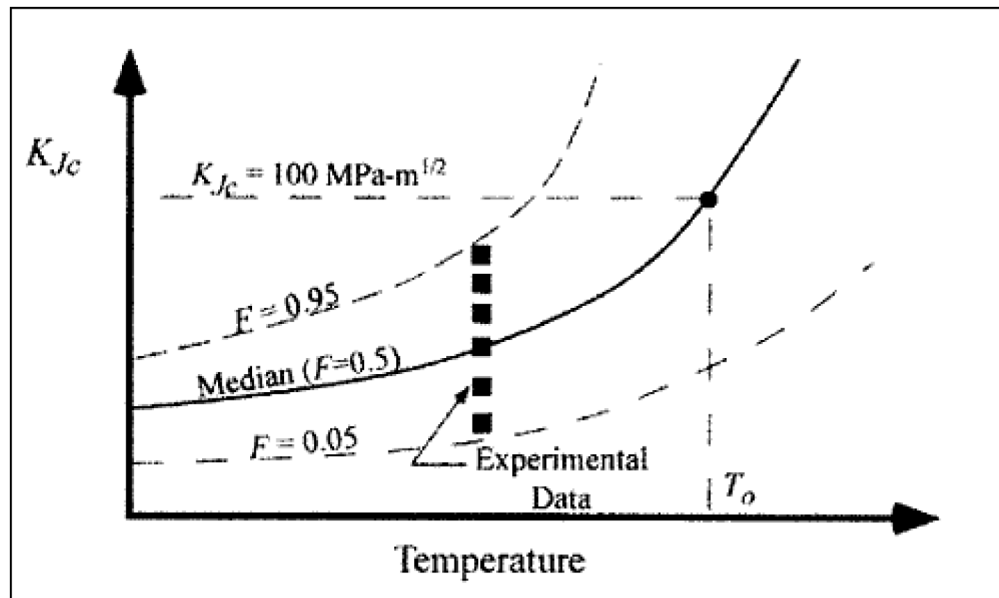


Figure (3c)—Fracture toughness master curve from American Society for Testing and Materials (ASTM) E1921



Figure (3d)—Flowchart explaining the use of the indenter method to calculate the fracture toughness using critical fracture stress and critical fracture strain models

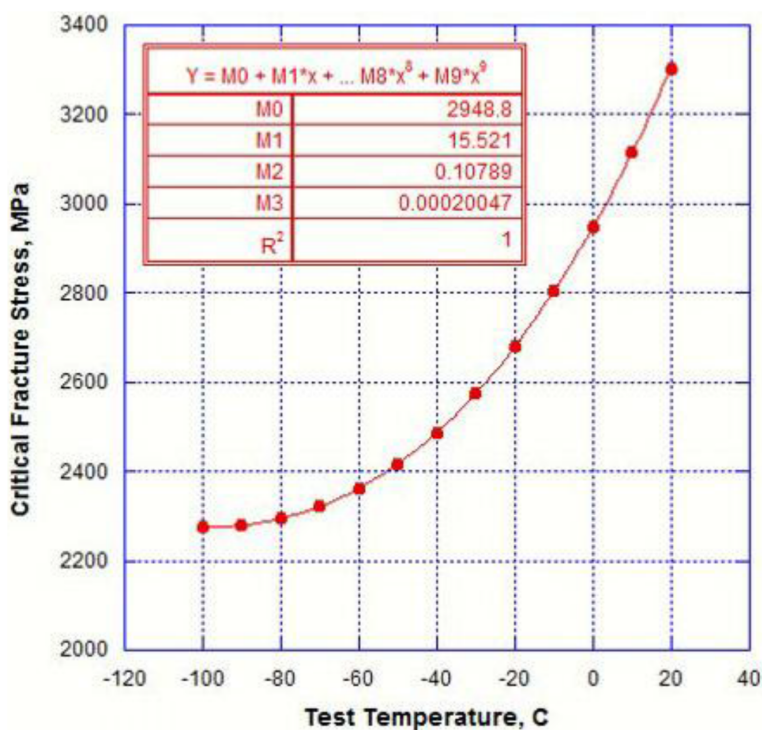
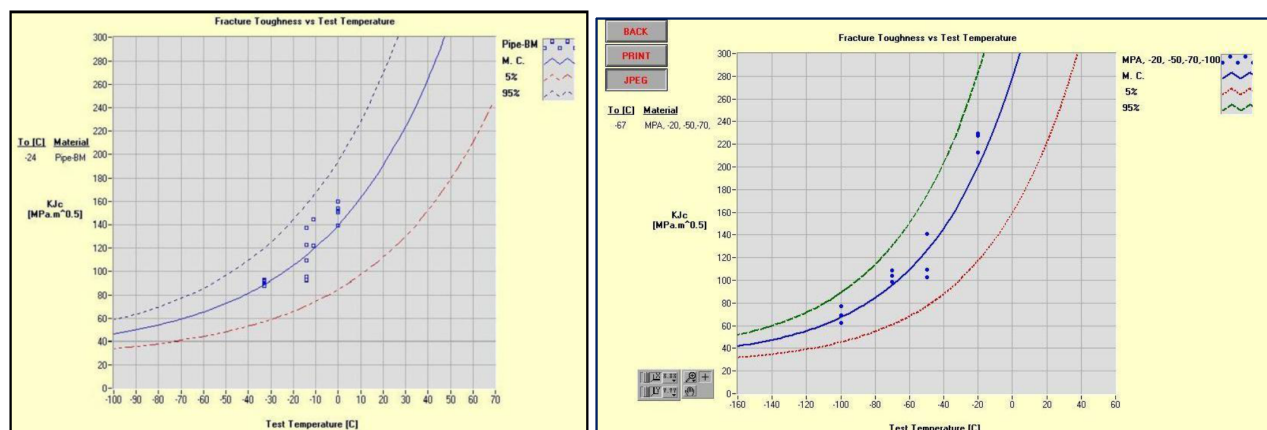


Figure 4—Critical fracture stress (sf) versus test temperature calculated from Equation (4)



(a) Pipeline steel

(b) Nuclear Pressure Vessel steel

Figure 5—Fracture toughness test results, their median master curves, and the 95% and 5% confidence curves.

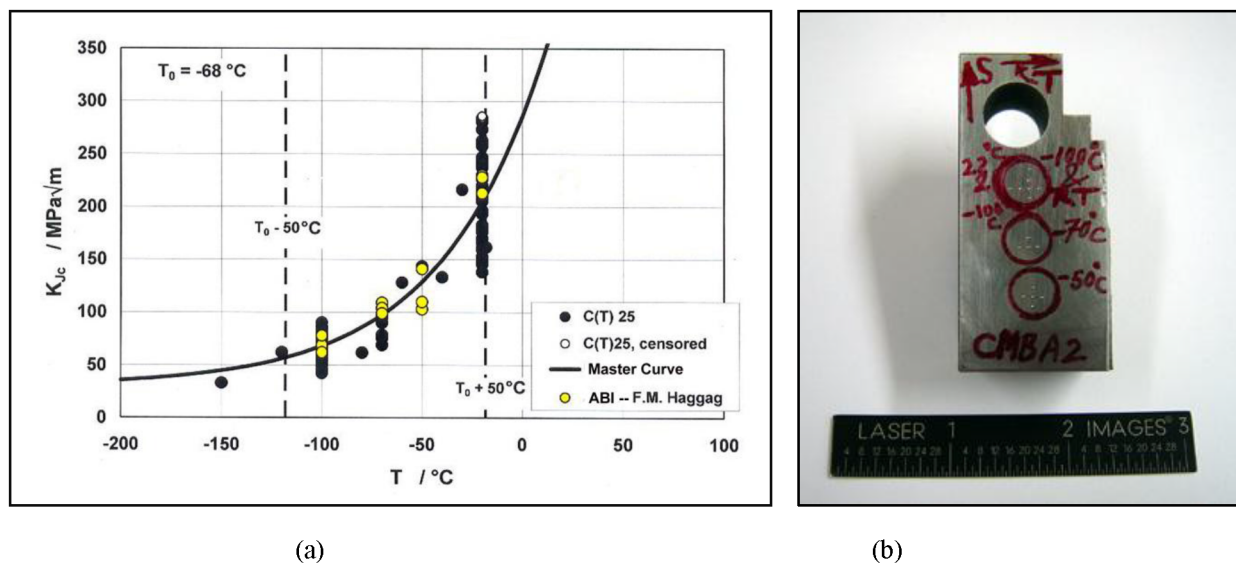


Figure 6—(a) Fracture toughness vs. temperature (b) showing Indentation test results and compact tension test results and indentation locations on the test specimen, courtesy of MPA University of Stuttgart, Germany due to the blind nature of all the Indentation tests.

The average Indentation-determined fracture toughness value from three room temperature tests on the nuclear pressure vessel sample of $221 \text{ MPa}\sqrt{\text{m}}$ was in good agreement with those from the destructive specimens with values ranging from $177 \text{ MPa}\sqrt{\text{m}}$ to $221 \text{ MPa}\sqrt{\text{m}}$. None of the room temperature test results from Indentation tests could be included in the Master curve because the critical fracture stress was not achieved at room temperature.

Testing the Outer Surface and the Reality of In-Service Conditions of Pipelines

The indentation test is conducted on the outer surface of the pipeline where the material is most vulnerable. This is important because this represents the area that is most vulnerable to cracking due to potential damage of the pipe coating material. Also, the outer surface is under manufacturing-tension (i.e., rolling a plate to a pipe), and since the indentation test being mostly compressive the data from the outer surface will be slightly lower/conservative. Although it tests a small volume of material, it interrogates several hundred grains of the pipe steel and is able to provide the bulk behavior of the test material.

Test Limitations

The ABI determined fracture toughness method described here is only applicable to carbon steels and it is not applicable to stainless steel. Additionally, the test requires a 63 RMS surface finish, so coatings or corrosion pitting must be removed before the test can be performed. This is essential because measurement is done to the micron level.

Case Study 1: Offshore Flare Boom Replacement

In early 2024, a deepwater facility was being reviewed for a life extension. During this review, it was noted that some piping materials within the flare boom served a dual purpose of both structure members and process piping, which required a full analysis for a low temperature service, such as a platform upset or flaring event. If the existing materials were found not to be Fit for Service, the entire flare boom structure would require a full replacement. Original plans estimated over \$20 million replacement cost, factoring in loss of production, offshore mobilization, logistics, fabrication, installation and commissioning of a

replacement flare boom. This made destructive testing an inefficient option for materials verification due to the cumulative losses created by system downtime required prior to any potential replacement project. In the case of non-destructive testing the best-case scenario, where the pipeline is FFS, would result in no replacement costs and no losses due to downtime.

The assessment included a positive material identification (PMI) and a mechanical integrity evaluation using the Automated Ball Indentation® (ABI®) method. By using both PMI and Indentation test methods, the material could first be confirmed with as-built documentation, and then most importantly, verify low temperature mechanical properties. The goal was to determine whether the existing flare boom structure met current fitness-for-service requirements and could remain in service with confidence in the event of a facility upset or extended flaring event.

Testing Scope and Methodology

- Tests were completed on Deepwater Flare boom structure—elevated, weathered, inaccessible without rope access or scaffolding.
- Full chemical analysis was completed through positive material identification (PMI).
- Yield strength, tensile strength, and fracture toughness values were obtained from Indenter tests as shown in Table 2 and plotted on the Master Curve as shown in Fig 7.
- Values were compared against the original design criteria as well as all possible process conditions and adverse weather conditions.

Key Findings

- The measured data revealed material properties matched the as built documentation. Most important were the fracture toughness results which confirmed the material's properties exceeded the minimum requirement, even at temperatures below what was originally reported on the material test report, despite visible surface weathering.
- The flare boom's structural integrity has more than adequate fracture toughness at the lowest anticipated service temperature was validated without a need for destructive testing or facility shutdown.

Table 2—Results of mechanical properties obtained from the Indenter tests

Test Name	Yield Par. A [MPa]	Yield Streng. (A,β) [MPa]	Eng. UTS [MPa]	Streng. Coeff(K) [MPa]	Strain Hard Exp. (n)	IDE [KJ/m ²]	KJc[MPa*m ^{0.5}]
24 pipe 1 t2c	1659	400	515	703	0.092	105.1	238.5
24 pipe 1 t4	1646	395	486	646	0.081	102.3	235.7
24 pipe 1 t5	1489	336	504	750	0.131	95.8	229.1
24 pipe 1 t6	1664	402	504	678	0.086	91.1	224.2
24 pipe 1 t7	1686	410	523	710	0.089	79.1	210.9
24 pipe 1 t8	1714	420	529	711	0.085	84.8	217.3

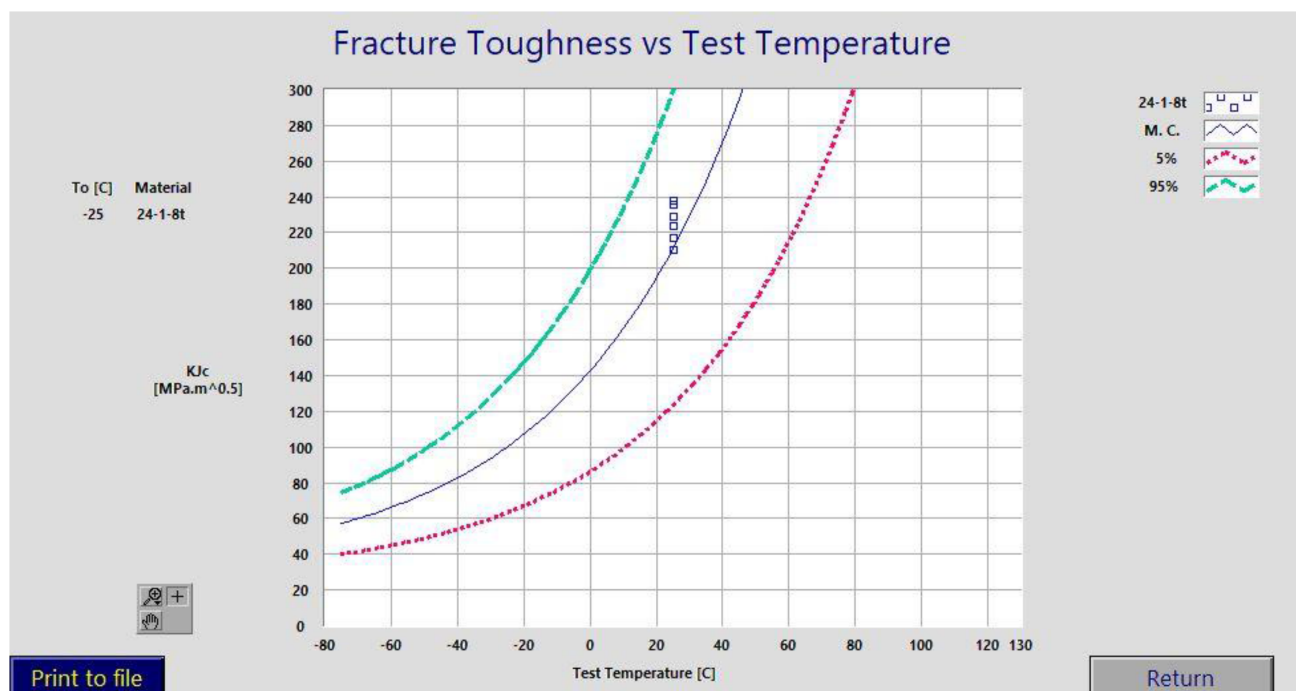


Figure 7—Fracture toughness vs. temperature Master curve showing Indenter test data

While not explicitly labelled on the Master Curve, data points occupy the range from 211.1 to 239.m^{0.5}. The multiple fracture toughness results at ambient temperature of each pipe location were used to calculate the nine conservative reference temperatures shown in Table 3. These reference temperatures were used in the Master Curve Equation 1 to calculate the fracture toughness values at -46°C (-50°F) as also shown in Table 3. Then the fracture toughness values at this low temperature were used in Equation F.66 of API-579 [$K_{Ic} = 8.47 (CVN)^{0.63}$, in the units of MPa $\sqrt{m}$ and J, respectively] to estimate the Charpy V-Notch (VCN) impact values as shown in Table 3. All the CVN values at -46°C (-50°F) are above the minimum required value of 20J, hence, all joints are fit for service. ***All estimated CVN values are higher than the minimum required value of 20J, hence, all nine joints are acceptable and fit for service.*** This was accomplished without any need to cut samples from any of the pipes.

Table 3—Results of the estimated Charpy V-Notch values from the fracture toughness (KJc) measured values

	To, C	KJc MPaM ^{1/2} , J	Charpy at -46C, J
24 Pipe #1	-25	77	33.2
24 Pipe #2	-30	81.7	36.5
24 Pipe #3	-27	78.8	34.5
24 Pipe #4	-28	79.7	35.1
24 x 30 Reducer	-30	81.7	36.5
30 Pipe #1	-33	84.7	38.6
30 x 24 Tee	-29	80.7	35.8
30 Pipe #2	-33	84.7	38.6
24 Pipe #5	-26	77.9	33.8

Outcome and Value Impact

The results of the testing proved the existing flare boom structure was Fit for Service, avoiding a \$20 million replacement project. This enabled reduced production downtime and safety risk due to elimination of an offshore construction project.

This case demonstrates the transformative value of integrating nondestructive mechanical testing into inspection workflows for offshore structures. The ability to quickly quantify mechanical properties non-destructively, in situ, working at heights and over the water, enabled a high-confidence decision with enormous cost and safety implications. As such, it serves as a compelling example of how modern material assessment tools can drive better asset integrity decisions, particularly in aging offshore infrastructure.

Case Study 2: 30" Flare Header

In order to meet new regulatory requirements, an aging gas plant was required to install additional instrumentation. It was necessary to add an instrument to an existing flare header for a gas plant to monitor emissions. To do this, a branch connection needs to be added to the existing 30" NPS flare line by using an in-service weld or hot tap. Prior to executing the work, multiple conditions were required to be verified. The original Material Test Reports (MTRs) for this flare line were unable to be verified. To confirm material properties, PMI using Laser Induced Breakdown Spectrometry (LIBS) was used to confirm the material's chemistry. In order to confirm mechanical properties, such as, yield, tensile or fracture toughness, it was necessary to obtain nondestructively to verify the material properties, before executing the work. The nondestructive mechanical testing method was used to very quickly confirm the existing material was able to be safely welded on during service with no impact on the line integrity and would to be fit for service.

Case Study 3: Cross-Verification of High Strength Casing Material Test Reports using nondestructive and traditional destructive mechanical testing.

Large deepwater high strength OTCG tubulars: it was important for acceptance and Mechanical Properties Identification and Verification (MPIV) and comparison of various mill's Certified Material Test Reports (cMTRs). Three different mills were compared to verify how close the cMTRs were to both nondestructive test results and destructive mechanical testing. data. This study evaluated three Q125-grade types [High collapse, or controlled yield] casing materials sourced from major global manufacturers—to verify mechanical property consistency and explore in-situ characterization via Indenter test. The goal was also to validate the Indenter test as a reliable alternative as non-destructive property verification to manufacturing standards and evaluate against destructive mechanical testing.

Estimating Charpy V-Notch (CVN) values from K_{Ic} Fracture Toughness

The API Fitness-For-Service Standard API 579-1 (ASME FFS-1) provides empirical equations (Annex F) for estimating static fracture toughness from impact energy CVN values over the three ranges of the lower shelf, transition region, and upper shelf using various equations as shown below. In this paper we demonstrate the estimation of CVN impact energy values from static fracture toughness results determined from the Indenter tests at ambient temperature.

For estimating the CVN values from nondestructive mechanical tests conducted at ambient/room temperature the following Ralf-Nowak-Barsom Equation F.60 (Annex F) is used for upper shelf correlation.

$$(K_{Ic}/YS)^2 = 0.64(CVN/YS-0.01) \quad (\text{MPa}\sqrt{\text{m}}, \text{MPa}, \text{ J}) \quad , \text{YS is yield strength} \quad \text{F.60}$$

For estimating the true fracture toughness from CVN impact energy in the transition region, the Sailors and Corton correlation is used.

$$K_{Jc} = 14.6 (\text{CVN})^{0.50} \quad (\text{MPa}\sqrt{\text{m}}, \text{J}) \quad \text{F.65}$$

For estimating CVN values from Indenter tests at room temperature, a conservative Reference Temperature (T_0) is determined from the Master Curve of ASTM E 1921, then this reference temperature is used to calculate the fracture toughness at the desired/specified low temperature, and finally the CVN values at the lower bound (lower shelf) is estimated from the following Equation (F.66).

$$K_{Jc} = 8.47 (\text{CVN})^{0.63} \quad (\text{MPa}\sqrt{\text{m}}, \text{MPa}, \text{J}) \quad \text{F.66}$$

Material Specifications and Test Samples

Each casing sample had a nominal wall thickness of 0.5 inches, with outer diameters ranging from 12 to 18 inches. All samples were designated Q125, a high-strength casing grade typically used in deep, high-pressure wells. Mill certificates reported the following properties:

Table 4—Mill Certificates for the Following Materials

Supplier	Yield Strength Avg (ksi) Average	Tensile Strength Avg (ksi)	Charpy Impact @ +32°F [0°C]
Material A	136.433	140.148	99.6 ft-lbs [135J]
Material B	127.134	142.15	119.5 ft-lb [162J]
Material C	145.3	155.6	87 ft-lb [118J]

Each sample underwent both Indenter testing and traditional destructive mechanical testing, including full-size Charpy V-notch (10x10 mm) on extracted coupons.

Comparison and Findings

Table 5—Comparison between ABI and Destructive Tests Methods for Yield and Tensile Strength – Materials A, B, and C

Material	Property	Destructive Test Avg	ABI Avg	Delta (%)
A	Yield Strength (ksi)	136.4	139.6	2.3%
B	Yield Strength (ksi)	134.7	129.4	-4%
C	Yield Strength (ksi)	145.3	141.8	-2.4%
A	Tensile Strength (ksi)	148.8	149.3	0.4%
B	Tensile Strength (ksi)	144.7	155.3	7.3%
C	Tensile Strength (ksi)	155.6	152.2	2.2%

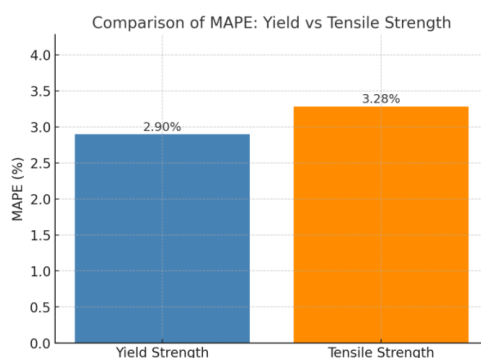


Figure 8—Mean Absolute Percentage Error (MAPE) for ABI and Destructive Tests for Yield and Tensile Strength – Materials A, B, and C

Table 6—Comparison CVN measured values and an estimation of CVN from KJc values using Equation F.65 for the transition region of CVN data. On Materials A, B, and C

Material	CVN Measured (J) at 0 °C	CVN Estimated from ABI (J) 0 °C	Difference	Percentage Difference	Notes
A	135	105.8	-29.2	-22%	Estimated CVN from Fracture Toughness by proxy via indentation fracture toughness using Equation F.65 for the transition region of CVN data
B	162	85.7	-76.3	-47%	
C	118	82.7	-35.3	-30%	

Ball Indenter Test derived fracture toughness values estimated using validated correlation models from load-depth curves. This test determined reference temperature (from eight ABI tests on each of the three casing materials was 5 °C for each material because they have very similar flow properties). The reference temperature was adjusted to -15 °C (the ASTM Standard E1921 states that if the reference temperatures are within 20 °C then the three casing materials are considered same material. Hence, for additional conservatism a 20 °C was added to the reference temperature of 5 °C). The new 25 °C reference temperature was used in the master curve to calculate the median fracture toughness value at 0 °C to estimate the CVN value using Equation F.65 of API 579-1 since all CVN tests were conducted at 0 °C (+32 °F).

Table 7—Comparison between ABI-measured Fracture Toughness and Fracture Toughness Estimated from CVN

Material	ABI Measured (KJc, MPa√m) at 20 °C	ABI Calculated (KJc, MPa√m) at 0 °C	Fracture Toughness Estimated from CVN Results (KJc, MPa√m) at 0 °C	Delta	Percentage Difference	Notes
A	151.0	93.7	128	-34.3	-26.8%	Estimated Fracture Toughness from CVN by linking $T_0 \approx T_{27J} - 18^\circ \text{C}$ and applying $K_{mat} \approx 0.54C_v + 55 \text{ (MPa}\sqrt{\text{m}}, C_v \text{ in J)}$ (Pisarski) ¹⁰
B	135.5	93.7	142	-48.3	-34.0%	
C	133.1	93.7	120	-26.3	-21.9%	

API 579-1/ASME-FFS-1 2021 Standard has changed from the 2007 version by eliminating the empirical equations for estimating the fracture toughness from the CVN results because of limited accuracy.

Since the MTR from the three vendors included only CVN data for testing at 0 °C (32 °F) we used the following Equations provided in Section 9F.2.4.2 of Annex 9F. The minimum specified yield strength (σ_{msys}) value for the 125Q casing material is 125 ksi or 862 MPa.

Using this σ_{msys} value in Equation 9F.22 the estimated value of the CVN at the upper shelf $C_v\text{-US}$ is 400 J. (295 ft-lbs). [$C_v\text{-US} = (0.290\sqrt{3} \sigma_{msys} - 27)^2 / (0.535 \sigma_{msys}) + (\sigma_{msys}/20) \text{ (J, MPa)}$] (9F.22) (9F

The CVN at the lower shelf is estimated to be 4.4 J from Equation 9F.24 [$C_v\text{-LS} = (K_{Id}\text{-LS}/14.155)^2$, J, MPa√m] and using $K_{Id}\text{-LS} = 29.7 \text{ MPa}\sqrt{\text{m}}$ (from Equation 9F.26)

Using the C and D values of 36.7 °C and -11.1 °C, respectively from Equations 9F.27 and 9F.28 for the CVN Reference Temperature Determination we obtain $T_{27Joules}$ using Equation 9F.32 to be $T_{27Joules} = -35.5^\circ \text{C}$ ($T_{20ft-lbs} = -31.8^\circ \text{F}$). [$T_{27Joules} = D + (C/4) \ln ((27 - C_v\text{-LS})/(C_v\text{-US}-27))^\circ \text{C, J}$].

Using the Equation from British Standard BS 7910 of estimating the Reference Temperature T_0 from Charpy estimates. $T_0 = T_{27Joules} - 18^\circ \text{C}$ (British Standard 7910).

Using this equation $T_0 = -35.5 - 18 = -53.5^\circ \text{C}$ in the ASTM Standard E 1921, produces a Median Fracture Toughness $KJc = 128.5 \text{ MPa}\sqrt{\text{m}}$ at the low temperature of -35.5 °C (at which the CVN values is 27 Joules or 20 ft-lbs).

We used σ_{msys} at 125 Ksi because it was slightly more conservative than what was directly measured by both ABI and the MTRs for all three materials.

For all the indentation results the reference temperature for all three materials was 5°C for each material, which is conservative higher value than -53.5°C explained above. Using this reference temperature in the Master Curve, the median fracture toughness value at 0°C is 93.7 MPa $\sqrt{m}$.

Key Observations

- Slight property variation between suppliers was observed, reinforcing the value of field validation—particularly for critical sour service or fracture-sensitive environments.
- The nondestructively obtained yield and tensile results closely matched destructive test values, showing differences of less than $\pm 7.3\%$.
- The selected method provided repeatable in-situ measurements within 30 minutes, compared to 10-15 business days required for destructive testing and machining of specimens.
- API 579-1/ASME FFS-1 2007 states "Charpy V-Notch impact data do not represent true fracture toughness data". The nature of the CVN test shows more scattering than true fracture toughness data. For the three vendors of the casing material the CVN values at 0°C varied from 118 to 162 Joules or a 37% variation while the ABI-determined fracture toughness results varied by 16% (133.1 to 151 MPa $\sqrt{m}$). Also, the true fracture toughness data allows the use of the fracture toughness Master Curve to calculate the median fracture toughness values at lower or higher temperatures than the test temperature.
- Applying Annex J of BS 7910:1999 to the CVN data in [Table 7](#) to estimate K_{Jc} shows a smaller difference between the measured K_{Jc} and the estimation of K_{Jc} from the CVN data, but it should be noted that estimating K_{Jc} from CVN is only meant to be an estimation and not a direct conversion because CVN is not a true measurement of fracture toughness.

This case study validates the utility of non-destructive, field-ready assessment of high-strength casing materials. The test can effectively confirm mill-supplied values and provide additional fracture toughness (K_{Jc}) results—critical for ensuring casing integrity, especially when materials are sourced globally and used in demanding operational settings. Indentation test represents a viable tool for compliance, quality control, and risk mitigation in modern casing programs.

Equations.

Fracture Toughness Calculation at Low Temperatures	$K_{Jc}(med) = 30 + 70e^{0.019(T-T_0)} \quad \text{MPa}\sqrt{\text{m}}$
Median Dynamic Fracture Toughness Calculation	$K_{Id}(med) = 30 + 70e^{0.019(T-[T_0+T_{shift}])} \quad \text{MPa} \sqrt{\text{m}}$
Transition Temperature/Barsom Correlation	$T_{shift} (^{\circ}F) = 215 - 1.5\sigma_{ys} (ksi)$ for 36 ksi < σ_{ys} < 140 ksi
Critical Fracture Stress Calculation	$\sigma_f = \sigma_y \left[1 + \ln \left(1 + 2360 \left(\frac{K_{Jc}}{\sigma_y} \right)^2 \right) \right] = 1.1 P_m$
CVN Impact Values Estimation in the Lower Shelf	KIc = 8.47 (CVN) ^{0.63} , in the units of MPa√m and J, respectively
CVN Impact Values Estimation in the Transition Temperature Region	(KIc/YS) ² = 0.64 (CVN/YS - 0.01) (MPa√m, MPa, J)

Conclusion

The above analysis and case studies demonstrate the viability of use of non-destructive mechanical properties testing to measure mechanical properties and fracture toughness data to use in assessing the fitness for service of existing equipment for continued service and/or repurposing. The case studies provide insights into multiple use cases for the Indenter test. A common theme is speed, flexibility, cost-effectiveness, and accuracy. Avoiding shutdown to extract samples for destructive testing allows generating large sets of data, more accurate statistical analyses and higher level of confidence in evaluating equipment integrity.

Glossary

ABI	Automated Ball Indentation
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
cMTR	certified MTR
CVN	Charpy Impact Energy/Charpy V-Notch
FFS	fitness-for-Service
HFTM	Haggag Fracture Toughness Method
LAST	Lowest Anticipated Service Temperature
LIBS	Laser Induced Breakdown Spectrometry
MDT	Minimum Design Temperature

MPIV	Mechanical Properties Identification and Verification
MTR	Material Test Reports
ORNL	Oak Ridge National Laboratory
PMI	Positive Material Identification
PRCI	Pipeline Research Council International

Nomenclature

D=	Indenter Diameter
d=	Indentation Chordal Diameter
K_{Jc} =	Static Fracture Toughness
K_{Id} =	Dynamic Fracture Toughness
IDE=	Indentation Deformation Energy
T_0 =	Reference Temperature Determined by the Statistical Analysis of ASTM Standard E 1921 as the Temperature at Fracture Toughness Level of 100 MPa $\sqrt{m}$
T=	Test Temperature
T_{shift} =	Transition Temperature for Dynamic Testing
YS/σ_{ys} =	Yield Strength
UTS/TS=	Ultimate Tensile Strength
CVN=	Charpy V-notch Impact Energy Value
σ_f =	Critical Fracture Stress
P_m =	Mean Pressure Below Spherical Indenter

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